

Homework 1

SI252 Reinforcement Learning

YiMing Li

Fall 2026

Problem 1. Random variables

A random variable turns the result of an experiment into a number, so we can calculate probabilities and averages. For example, let X be the number of heads in two coin tosses:

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

X is a function because it assigns a number to each outcome. The rule is fixed, but we do not know which outcome will occur, so the value of X is uncertain.

Problem 2. Probability and estimation identities

(a)

1. Both X and Y are discrete

By the definition of conditional probability,

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}.$$

Also,

$$P(X = x, Y = y) = P(X = x \mid Y = y)P(Y = y).$$

Substituting gives

$$P(Y = y \mid X = x) = \frac{P(X = x \mid Y = y)P(Y = y)}{P(X = x)}.$$

2. X is discrete and Y is continuous

Fix x such that $P(X = x) > 0$. For every measurable set $B \subseteq \mathbb{R}$, conditioning on Y yields

$$P(X = x, Y \in B) = \int_B P(X = x \mid Y = y) f_Y(y) dy.$$

Hence,

$$\begin{aligned} P(Y \in B \mid X = x) &= \frac{P(X = x, Y \in B)}{P(X = x)} \\ &= \int_B \frac{P(X = x \mid Y = y) f_Y(y)}{P(X = x)} dy. \end{aligned}$$

By the definition of a conditional density, the integrand is a density of Y conditional on $X = x$. Therefore,

$$f_{Y|X}(y | x) = \frac{P(X = x | Y = y)f_Y(y)}{P(X = x)}.$$

3. X is continuous and Y is discrete

Fix y such that $P(Y = y) > 0$. For every measurable set $A \subseteq \mathbb{R}$,

$$\begin{aligned} P(X \in A, Y = y) &= P(Y = y)P(X \in A | Y = y) \\ &= \int_A f_{X|Y}(x | y)P(Y = y) dx. \end{aligned}$$

On the other hand, conditioning on X gives

$$P(X \in A, Y = y) = \int_A P(Y = y | X = x)f_X(x) dx.$$

Since these integrals are equal for every measurable set A , their integrands are equal almost everywhere:

$$P(Y = y | X = x)f_X(x) = f_{X|Y}(x | y)P(Y = y).$$

Dividing by $f_X(x) > 0$, we obtain

$$P(Y = y | X = x) = \frac{f_{X|Y}(x | y)P(Y = y)}{f_X(x)}.$$

4. Both X and Y are continuous and admit a joint density

Let $f_{X,Y}(x, y)$ denote the joint density. By the definition of conditional density,

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

Moreover,

$$f_{X,Y}(x, y) = f_{X|Y}(x | y)f_Y(y).$$

Substituting yields

$$f_{Y|X}(y | x) = \frac{f_{X|Y}(x | y)f_Y(y)}{f_X(x)}.$$

(b)

1. Both X and Y are discrete

The events $\{Y = y\}$, taken over the possible values of Y , form a countable partition of the sample space up to a null set. Therefore, by countable additivity,

$$P(X = x) = \sum_y P(X = x, Y = y).$$

Using the definition of conditional probability,

$$P(X = x, Y = y) = P(X = x | Y = y)P(Y = y).$$

Consequently,

$$P(X = x) = \sum_y P(X = x | Y = y)P(Y = y).$$

The sum may be restricted to values y for which $P(Y = y) > 0$.

2. X is discrete and Y is continuous

Fix x such that $P(X = x) > 0$. By the previously established Bayes' rule,

$$f_{Y|X}(y | x) = \frac{P(X = x | Y = y)f_Y(y)}{P(X = x)}.$$

Rearranging gives

$$P(X = x | Y = y)f_Y(y) = P(X = x)f_{Y|X}(y | x).$$

Integrating both sides with respect to y , we obtain

$$\begin{aligned} \int_{-\infty}^{\infty} P(X = x | Y = y)f_Y(y) dy &= P(X = x) \int_{-\infty}^{\infty} f_{Y|X}(y | x) dy \\ &= P(X = x). \end{aligned}$$

3. X is continuous and Y is discrete

Let $A \subseteq \mathbb{R}$ be any measurable set. Applying the discrete law of total probability to the event $\{X \in A\}$, we have

$$P(X \in A) = \sum_y P(X \in A | Y = y)P(Y = y).$$

By the definition of conditional density,

$$P(X \in A | Y = y) = \int_A f_{X|Y}(x | y) dx.$$

Thus,

$$\begin{aligned} P(X \in A) &= \sum_y \left[\int_A f_{X|Y}(x | y) dx \right] P(Y = y) \\ &= \int_A \left[\sum_y f_{X|Y}(x | y)P(Y = y) \right] dx. \end{aligned}$$

The interchange of the sum and the integral is justified by Tonelli's theorem, since all the terms are nonnegative.

On the other hand,

$$P(X \in A) = \int_A f_X(x) dx.$$

Since these expressions agree for every measurable set A , uniqueness of densities yields

$$f_X(x) = \sum_y f_{X|Y}(x | y)P(Y = y).$$

4. Both X and Y are continuous and admit a joint density

Assume that X and Y have a joint density $f_{X,Y}(x, y)$. By marginalization,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

The conditional density satisfies

$$f_{X,Y}(x, y) = f_{X|Y}(x | y) f_Y(y)$$

almost everywhere. Substituting gives

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x | y) f_Y(y) dy.$$

(c)

Assume that $E[Y^2] < \infty$.

$$\text{Var}(Y | X) = E[Y^2 | X] - (E[Y | X])^2.$$

Taking expectations and applying the law of total expectation gives

$$\begin{aligned} E[\text{Var}(Y | X)] &= E[E[Y^2 | X]] - E[(E[Y | X])^2] \\ &= E[Y^2] - E[(E[Y | X])^2]. \end{aligned}$$

Using the variance identity $\text{Var}(Z) = E[Z^2] - (E[Z])^2$, we obtain

$$\begin{aligned} \text{Var}(E[Y | X]) &= E[(E[Y | X])^2] - (E[E[Y | X]])^2 \\ &= E[(E[Y | X])^2] - (E[Y])^2, \end{aligned}$$

where the last equality follows from the law of total expectation.

Adding the expressions above, the middle terms cancel:

$$\begin{aligned} &E[\text{Var}(Y | X)] + \text{Var}(E[Y | X]) \\ &= E[Y^2] - E[(E[Y | X])^2] + E[(E[Y | X])^2] - (E[Y])^2 \\ &= E[Y^2] - (E[Y])^2 \\ &= \text{Var}(Y). \end{aligned}$$

Therefore,

$$\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E[Y | X]).$$

(d)

Assume that $E[X^2] < \infty$, $E[Y^2] < \infty$, and $\text{Var}(X) > 0$. We seek constants a and b that minimize

$$Q(a, b) = E[(Y - a - bX)^2].$$

$$Q(a, b) = E[Y^2] - 2aE[Y] - 2bE[XY] + a^2 + 2abE[X] + b^2E[X^2].$$

$$\frac{\partial Q}{\partial a} = -2E[Y] + 2a + 2bE[X] = 0.$$

Hence,

$$a = E[Y] - bE[X].$$

Similarly,

$$\frac{\partial Q}{\partial b} = -2E[XY] + 2aE[X] + 2bE[X^2] = 0,$$

so

$$E[XY] = aE[X] + bE[X^2].$$

Substituting $a = E[Y] - bE[X]$, we obtain

$$E[XY] = E[X]E[Y] - b(E[X])^2 + bE[X^2].$$

Rearranging,

$$E[XY] - E[X]E[Y] = b(E[X^2] - (E[X])^2).$$

By the definitions of covariance and variance,

$$\text{Cov}(X, Y) = b \text{Var}(X).$$

Therefore,

$$b = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

Thus,

$$L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X]).$$

The objective is a convex quadratic in (a, b) , with a positive-definite Hessian when $\text{Var}(X) > 0$, so this stationary point is the unique minimum. If $\text{Var}(X) = 0$, X is constant almost surely and the best predicted value is simply $E[Y]$.

(e)

Let $m(X) = \mathbb{E}[Y | X]$, and assume that Y has a finite second moment. For every measurable ϕ with $\mathbb{E}[\phi(X)^2] < \infty$, the tower property gives

$$\begin{aligned} \mathbb{E}[(Y - m(X))\phi(X)] &= \mathbb{E}[\mathbb{E}[(Y - m(X))\phi(X) | X]] \\ &= \mathbb{E}[\phi(X)(\mathbb{E}[Y | X] - m(X))] \\ &= 0. \end{aligned}$$

This proves the orthogonality property.

Conversely, assume that $g(X)$ has a finite second moment and that, for every such measurable function ϕ ,

$$\mathbb{E}[(Y - g(X))\phi(X)] = 0.$$

By the orthogonality property just proved,

$$\mathbb{E}[(Y - m(X))\phi(X)] = 0.$$

Subtracting the second equality from the first gives

$$\mathbb{E}[(m(X) - g(X))\phi(X)] = 0.$$

Since this holds for every such ϕ , we may choose

$$\phi(X) = m(X) - g(X),$$

which is square-integrable. Therefore,

$$\mathbb{E}[(m(X) - g(X))^2] = 0.$$

Since a nonnegative random variable with zero expectation is zero almost surely, $g(X) = m(X) = \mathbb{E}[Y \mid X]$ almost surely.

(f)

Proof. Let $\mu = \mathbb{E}[Z]$. By Chebyshev's inequality,

$$\mathbb{P}(|Z - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$$

We consider two cases:

- If $a^2 \leq \sigma^2$, then

$$\mathbb{P}(|Z - \mu| \geq a) \leq 1 \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

- If $a^2 > \sigma^2$, then $\sigma^2 + a^2 \leq 2a^2$, and hence

$$\mathbb{P}(|Z - \mu| \geq a) \leq \frac{\sigma^2}{a^2} \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

Therefore, for every $a > 0$,

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq a) \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

(g)

Proof. Let

$$Y_k = X_k - \mathbb{E}[X_k], \quad V = \sum_{k=1}^n (b_k - a_k)^2.$$

Then the Y_k are independent, $\mathbb{E}[Y_k] = 0$, and

$$a_k - \mathbb{E}[X_k] \leq Y_k \leq b_k - \mathbb{E}[X_k].$$

Thus, by Hoeffding's lemma, for every $\lambda > 0$,

$$\mathbb{E}[e^{\lambda Y_k}] \leq \exp\left(\frac{\lambda^2(b_k - a_k)^2}{8}\right).$$

Using independence, we obtain

$$\begin{aligned}\mathbb{E}[e^{\lambda(S_n - \mu)}] &= \prod_{k=1}^n \mathbb{E}[e^{\lambda Y_k}] \\ &\leq \exp\left(\frac{\lambda^2 V}{8}\right).\end{aligned}$$

Assume $V > 0$ and $t > 0$. By Markov's inequality, for every $\lambda > 0$,

$$\begin{aligned}\mathbb{P}(S_n - \mu \geq t) &= \mathbb{P}(e^{\lambda(S_n - \mu)} \geq e^{\lambda t}) \\ &\leq e^{-\lambda t} \mathbb{E}[e^{\lambda(S_n - \mu)}] \\ &\leq \exp\left(-\lambda t + \frac{\lambda^2 V}{8}\right).\end{aligned}$$

Choosing $\lambda = 4t/V$ gives

$$\mathbb{P}(S_n - \mu \geq t) \leq \exp\left(-\frac{2t^2}{V}\right).$$

Applying the same argument to the independent random variables $-X_1, \dots, -X_n$, whose interval lengths remain $b_k - a_k$, yields

$$\mathbb{P}(S_n - \mu \leq -t) = \mathbb{P}(\mu - S_n \geq t) \leq \exp\left(-\frac{2t^2}{V}\right).$$

Therefore, by the union bound,

$$\begin{aligned}\mathbb{P}(|S_n - \mu| \geq t) &\leq \mathbb{P}(S_n - \mu \geq t) + \mathbb{P}(S_n - \mu \leq -t) \\ &\leq 2 \exp\left(-\frac{2t^2}{V}\right).\end{aligned}$$

For $t = 0$, the inequality holds trivially since $1 \leq 2$. Hence, when $V > 0$,

$$\mathbb{P}(|S_n - \mu| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2}\right), \quad t \geq 0.$$

If $V = 0$, all X_k are constant almost surely, so $S_n = \mu$ almost surely. In this degenerate case, the tail probability is 0 for $t > 0$ and 1 for $t = 0$.

Problem 3. Exponential conditional expectations

Assume that $X_i \sim \text{Exp}(\lambda_i)$, where λ_i is the rate parameter. Then

$$f_{X_i}(x) = \lambda_i e^{-\lambda_i x}, \quad x > 0.$$

By the memoryless property of the exponential distribution,

$$X_i - t \mid X_i > t \sim \text{Exp}(\lambda_i).$$

Therefore,

$$E(X_1 \mid X_1 > 2026) = 2026 + E(X_1) = \boxed{2026 + \frac{1}{\lambda_1}}.$$

For the second term,

$$E(X_1 \mid X_1 < 2026) = \frac{\int_0^{2026} x \lambda_1 e^{-\lambda_1 x} dx}{P(X_1 < 2026)}.$$

Evaluating the integral gives

$$E(X_1 \mid X_1 < 2026) = \frac{1}{\lambda_1} - \frac{2026 e^{-2026 \lambda_1}}{1 - e^{-2026 \lambda_1}}$$

or, equivalently,

$$E(X_1 \mid X_1 < 2026) = \frac{1}{\lambda_1} - \frac{2026}{e^{2026 \lambda_1} - 1}$$

Finally, since X_1, X_2, X_3 are independent, we can apply the memoryless property to each variable:

$$\begin{aligned} E(X_1 + X_2 + X_3 \mid X_1 > 1997, X_2 > 2014, X_3 > 2026) \\ = \left(1997 + \frac{1}{\lambda_1}\right) + \left(2014 + \frac{1}{\lambda_2}\right) + \left(2026 + \frac{1}{\lambda_3}\right). \end{aligned}$$

Hence,

$$E(X_1 + X_2 + X_3 \mid X_1 > 1997, X_2 > 2014, X_3 > 2026) = 6037 + \frac{1}{\lambda_1} + \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$$

Problem 4. Bivariate normal construction

Let Z_1, Z_2 be independent standard normal random variables, and define

$$X = \sigma_X Z_1 + \mu_X, \quad Y = \sigma_Y \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right) + \mu_Y,$$

where $\sigma_X, \sigma_Y > 0$ and $-1 < \rho < 1$.

(a) For any $a, b \in \mathbb{R}$,

$$aX + bY = a\mu_X + b\mu_Y + (a\sigma_X + b\rho\sigma_Y)Z_1 + b\sigma_Y\sqrt{1 - \rho^2}Z_2.$$

Since a linear combination of independent normal random variables is normal, $aX + bY$ is normal. Therefore, X and Y are jointly normal.

(b) We have

$$\text{Var}(X) = \sigma_X^2, \quad \text{Var}(Y) = \sigma_Y^2(\rho^2 + 1 - \rho^2) = \sigma_Y^2.$$

Using independence,

$$\begin{aligned} \text{Cov}(X, Y) &= \sigma_X \sigma_Y \left[\rho \text{Var}(Z_1) + \sqrt{1 - \rho^2} \text{Cov}(Z_1, Z_2) \right] \\ &= \rho \sigma_X \sigma_Y. \end{aligned}$$

Hence,

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \rho$$

(c) The inverse transformation is

$$z_1 = \frac{x - \mu_X}{\sigma_X}, \quad z_2 = \frac{\frac{y - \mu_Y}{\sigma_Y} - \rho \frac{x - \mu_X}{\sigma_X}}{\sqrt{1 - \rho^2}}.$$

Its Jacobian determinant has absolute value

$$\left| \frac{\partial(z_1, z_2)}{\partial(x, y)} \right| = \frac{1}{\sigma_X \sigma_Y \sqrt{1 - \rho^2}}.$$

Since

$$f_{Z_1, Z_2}(z_1, z_2) = \frac{1}{2\pi} \exp\left(-\frac{z_1^2 + z_2^2}{2}\right),$$

the change-of-variables formula gives

$$\begin{aligned} f_{X, Y}(x, y) &= \frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} \\ &\times \exp\left\{-\frac{1}{2(1 - \rho^2)} \left[\frac{(x - \mu_X)^2}{\sigma_X^2} - \frac{2\rho(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2} \right]\right\}, \end{aligned}$$

for all $x, y \in \mathbb{R}$.

Problem 5. A three-state Markov chain

The transition matrix, with states ordered as 1, 2, 3, is

$$P = \begin{pmatrix} \frac{1}{4} & 0 & \frac{3}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix},$$

where

$$p_{ij} = P(X_{n+1} = j \mid X_n = i).$$

(a)

Since the chain is time-homogeneous, the one-step transition probabilities do not depend on time. Therefore,

$$P(X_5 = 3 \mid X_4 = 2) = p_{23} = \frac{1}{2},$$

and

$$P(X_4 = 1 \mid X_3 = 2) = p_{21} = \frac{1}{2}.$$

(b)

Let $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3)$ be the stationary distribution. It satisfies

$$\boldsymbol{\pi}P = \boldsymbol{\pi}, \quad \pi_1 + \pi_2 + \pi_3 = 1.$$

Thus,

$$\begin{cases} \pi_1 = \frac{1}{4}\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{2}\pi_3, \\ \pi_2 = \frac{1}{4}\pi_3, \\ \pi_3 = \frac{3}{4}\pi_1 + \frac{1}{2}\pi_2 + \frac{1}{4}\pi_3. \end{cases}$$

Substituting $\pi_2 = \frac{1}{4}\pi_3$ into the first equation gives

$$\frac{3}{4}\pi_1 = \frac{1}{8}\pi_3 + \frac{1}{2}\pi_3 = \frac{5}{8}\pi_3,$$

so

$$\pi_1 = \frac{5}{6}\pi_3.$$

Using the normalization condition,

$$\left(\frac{5}{6} + \frac{1}{4} + 1\right)\pi_3 = 1,$$

which yields

$$\pi_3 = \frac{12}{25}, \quad \pi_2 = \frac{3}{25}, \quad \pi_1 = \frac{2}{5}.$$

Hence,

$$\boldsymbol{\pi} = \left(\frac{2}{5}, \frac{3}{25}, \frac{12}{25}\right).$$

(c)

The period of state i is defined by

$$d(i) = \gcd\{n \geq 1 : (P^n)_{ii} > 0\}.$$

Starting from state 2, the following return paths have positive probability:

$$2 \rightarrow 3 \rightarrow 2 \quad \text{and} \quad 2 \rightarrow 3 \rightarrow 3 \rightarrow 2.$$

Thus, returns are possible in both two and three steps. Since $\gcd(2, 3) = 1$,

$$d(2) = 1.$$

All three states communicate, so the chain is irreducible and all states have the same period. Therefore, the period of the whole chain is 1.

(d)

By the multiplication rule and the Markov property,

$$\begin{aligned} P(X_0 = 3, X_1 = 2, X_2 = 1) &= P(X_0 = 3) P(X_1 = 2 \mid X_0 = 3) P(X_2 = 1 \mid X_1 = 2) \\ &= \frac{1}{5} \cdot \frac{1}{4} \cdot \frac{1}{2} \\ &= \frac{1}{40}. \end{aligned}$$

(e)

Since there are two transitions from time 3 to time 5,

$$P(X_5 = j \mid X_3 = 1) = (P^2)_{1j} = \sum_{k=1}^3 p_{1k} p_{kj}.$$

Consequently,

$$P(X_5 = 1 \mid X_3 = 1) = \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{7}{16},$$

$$P(X_5 = 2 \mid X_3 = 1) = \frac{1}{4} \cdot 0 + \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16},$$

$$P(X_5 = 3 \mid X_3 = 1) = \frac{1}{4} \cdot \frac{3}{4} + \frac{3}{4} \cdot \frac{1}{4} = \frac{6}{16}.$$

The conditional expectation is

$$\begin{aligned} E(X_5 \mid X_3 = 1) &= 1 \cdot \frac{7}{16} + 2 \cdot \frac{3}{16} + 3 \cdot \frac{6}{16} \\ &= \frac{31}{16}. \end{aligned}$$

The conditional second moment is

$$\begin{aligned} E(X_5^2 \mid X_3 = 1) &= 1^2 \cdot \frac{7}{16} + 2^2 \cdot \frac{3}{16} + 3^2 \cdot \frac{6}{16} \\ &= \frac{73}{16}. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Var}(X_5 \mid X_3 = 1) &= E(X_5^2 \mid X_3 = 1) - [E(X_5 \mid X_3 = 1)]^2 \\ &= \frac{73}{16} - \left(\frac{31}{16}\right)^2 \\ &= \frac{207}{256}. \end{aligned}$$

Problem 6. Estimating a coin's head probability

Assume $n \geq 1$. Given p , the number of heads satisfies

$$X \mid p \sim \text{Binomial}(n, p).$$

Hence, for the observation $X = k$,

$$P(X = k \mid p) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \leq k \leq n.$$

(a) Maximum Likelihood Estimation (MLE)

Treating p as an unknown constant, the likelihood function is

$$L(p) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \leq p \leq 1.$$

For $0 < k < n$, the log-likelihood is

$$\ell(p) = \log \binom{n}{k} + k \log p + (n-k) \log(1-p).$$

Differentiating and setting the derivative equal to zero gives

$$\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0,$$

so

$$k(1-p) = (n-k)p \implies p = \frac{k}{n}.$$

Moreover,

$$\ell''(p) = -\frac{k}{p^2} - \frac{n-k}{(1-p)^2} < 0,$$

so this point maximizes the likelihood.

If $k = 0$, the likelihood is maximized at $p = 0$; if $k = n$, it is maximized at $p = 1$. Thus, in all cases,

$$\hat{p}_{\text{MLE}} = \frac{k}{n}.$$

(b) Maximum A Posteriori Estimation (MAP)

The prior distribution is

$$p \sim \text{Beta}(1, 1),$$

which is the uniform distribution on $[0, 1]$. Therefore, the prior density is $f(p) = 1$ on this interval.

By Bayes' rule,

$$f(p \mid X = k) \propto P(X = k \mid p) f(p) \propto p^k (1-p)^{n-k}.$$

Consequently,

$$p \mid X = k \sim \text{Beta}(k+1, n-k+1).$$

The MAP estimate maximizes the posterior density. Since the prior density is constant, maximizing the posterior density is equivalent to maximizing the likelihood. Hence,

$$\hat{p}_{\text{MAP}} = \frac{k}{n},$$

including the boundary cases $k = 0$ and $k = n$.

(c) Minimum Mean Squared Error Estimation (MMSE)

The MMSE estimate is the posterior mean:

$$\hat{p}_{\text{MMSE}} = E[p \mid X = k].$$

Since

$$\text{Unif}(0, 1) = \text{Beta}(1, 1),$$

the posterior distribution is again

$$p \mid X = k \sim \text{Beta}(k + 1, n - k + 1).$$

The mean of a $\text{Beta}(\alpha, \beta)$ distribution is $\alpha/(\alpha + \beta)$. Therefore,

$$\hat{p}_{\text{MMSE}} = \frac{k + 1}{n + 2}.$$

(d) Linear Least Squares Estimation (LLSE)

Using the given formula with $Y = p$, we obtain

$$\hat{p}_{\text{LLSE}} = E[p] + \frac{\text{Cov}(X, p)}{\text{Var}(X)}(k - E[X]).$$

All moments below are computed under the joint distribution specified by the prior and the observation model.

Since $p \sim \text{Unif}(0, 1)$,

$$E[p] = \frac{1}{2}, \quad E[p^2] = \frac{1}{3}, \quad \text{Var}(p) = \frac{1}{12}.$$

Also, $E[X \mid p] = np$, so the law of total expectation gives

$$E[X] = E[E[X \mid p]] = nE[p] = \frac{n}{2}.$$

By conditioning on p ,

$$E[Xp] = E[pE[X \mid p]] = nE[p^2] = \frac{n}{3}.$$

Thus,

$$\text{Cov}(X, p) = E[Xp] - E[X]E[p] = \frac{n}{3} - \frac{n}{2} \cdot \frac{1}{2} = \frac{n}{12}.$$

By the law of total variance,

$$\begin{aligned} \text{Var}(X) &= E[\text{Var}(X \mid p)] + \text{Var}(E[X \mid p]) \\ &= E[np(1 - p)] + \text{Var}(np) \\ &= n \left(\frac{1}{2} - \frac{1}{3} \right) + n^2 \left(\frac{1}{12} \right) \\ &= \frac{n(n + 2)}{12}. \end{aligned}$$

Therefore,

$$\begin{aligned}\hat{p}_{\text{LLSE}} &= \frac{1}{2} + \frac{n/12}{n(n+2)/12} \left(k - \frac{n}{2}\right) \\ &= \frac{1}{2} + \frac{k - n/2}{n+2} \\ &= \frac{k+1}{n+2}.\end{aligned}$$

Problem 7. Gaussian MMSE and LLSE

Method 1

Let

$$\mu_X = E[X], \quad \mu_Y = E[Y], \quad a = \frac{\text{Cov}(X, Y)}{\text{Var}(X)},$$

where we assume that $\text{Var}(X) > 0$. Define

$$Z = Y - \mu_Y - a(X - \mu_X).$$

Then

$$Y = \mu_Y + a(X - \mu_X) + Z.$$

First, the residual Z has mean zero:

$$\begin{aligned}E[Z] &= E[Y] - \mu_Y - a(E[X] - \mu_X) \\ &= 0.\end{aligned}$$

Moreover,

$$\begin{aligned}\text{Cov}(X, Z) &= E[(X - E[X])(Z - E[Z])] \\ &= E[(X - \mu_X)Z] \\ &= E[(X - \mu_X)((Y - \mu_Y) - a(X - \mu_X))] \\ &= E[(X - \mu_X)(Y - \mu_Y)] - aE[(X - \mu_X)^2] \\ &= \text{Cov}(X, Y) - a \text{Var}(X) \\ &= 0.\end{aligned}$$

Since X and Y are jointly Gaussian and Z is an affine function of X and Y , the variables X and Z are also jointly Gaussian. Thus, their zero covariance implies that they are independent. Consequently,

$$E[Z | X] = E[Z] = 0.$$

Taking conditional expectations gives

$$\begin{aligned}E[Y | X] &= E[\mu_Y + a(X - \mu_X) + Z | X] \\ &= \mu_Y + a(X - \mu_X) + E[Z | X] \\ &= \mu_Y + a(X - \mu_X).\end{aligned}$$

To verify the minimum mean squared error property, write

$$m(X) = \mu_Y + a(X - \mu_X).$$

Since $Y = m(X) + Z$, the mean squared error of $m(X)$ is

$$E[(Y - m(X))^2] = E[Z^2].$$

Any other square-integrable predictor can be written as

$$g(X) = m(X) + d(X).$$

Using $E[Z | X] = 0$, we obtain

$$E[Zd(X)] = E[d(X)E[Z | X]] = 0.$$

Therefore,

$$\begin{aligned} E[(Y - g(X))^2] &= E[(Z - d(X))^2] \\ &= E[Z^2] - 2E[Zd(X)] + E[d(X)^2] \\ &= E[Z^2] + E[d(X)^2] \\ &\geq E[Z^2]. \end{aligned}$$

Hence $m(X) = E[Y | X]$ is the MMSE predictor. Since $m(X)$ is also affine in X , it is the best linear least-squares predictor $L[Y | X]$. Thus,

$$E[Y | X] = L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X])$$

Method 2

Let

$$\mu_X = E[X], \quad \mu_Y = E[Y], \quad \sigma_X^2 = \text{Var}(X) > 0, \quad \sigma_Y^2 = \text{Var}(Y).$$

First assume $\sigma_Y > 0$. Define

$$U = \frac{X - \mu_X}{\sigma_X}, \quad V = \frac{Y - \mu_Y}{\sigma_Y}, \quad \rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

For $|\rho| < 1$, the joint density of U, V is

$$f_{U,V}(u, v) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{u^2 - 2\rho uv + v^2}{2(1-\rho^2)}\right).$$

Given $U = u$, treat u as a constant. Completing the square in v , we get

$$u^2 - 2\rho uv + v^2 = (v - \rho u)^2 + (1 - \rho^2)u^2.$$

Therefore,

$$f_{U,V}(u, v) = \frac{e^{-u^2/2}}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{(v - \rho u)^2}{2(1-\rho^2)}\right).$$

Since $f_{V|U}(v | u) = f_{U,V}(u, v)/f_U(u)$, all factors independent of v can be absorbed into a constant. Thus,

$$f_{V|U}(v | u) \propto \exp\left(-\frac{(v - \rho u)^2}{2(1 - \rho^2)}\right).$$

This is a Gaussian density symmetric about $v = \rho u$. Hence its mean is

$$E[V | U = u] = \rho u.$$

To return to the original variables, note that

$$Y = \mu_Y + \sigma_Y V.$$

Also, $X = x$ is equivalent to $U = (x - \mu_X)/\sigma_X$. Therefore,

$$\begin{aligned} E[Y | X = x] &= \mu_Y + \sigma_Y E[V | X = x] \\ &= \mu_Y + \sigma_Y E\left[V \mid U = \frac{x - \mu_X}{\sigma_X}\right] \\ &= \mu_Y + \sigma_Y \rho \frac{x - \mu_X}{\sigma_X} \\ &= \mu_Y + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(x - \mu_X). \end{aligned}$$

The conditional expectation minimizes mean squared error among all predictors based on X . Since the expression above is affine in X , it is also the best linear predictor $L[Y | X]$. Thus,

$$E[Y | X] = L[Y | X] = E[Y] + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E[X]).$$

Degenerate cases. If $\sigma_Y = 0$, then $Y = \mu_Y$ almost surely and the result is immediate. If $\sigma_X, \sigma_Y > 0$ and $|\rho| = 1$, then

$$\text{Var}(V - \rho U) = 1 - \rho^2 = 0,$$

so $V = \rho U$ almost surely, giving the same affine conditional mean. If $\text{Var}(X) = 0$, the displayed covariance ratio is undefined, but both the MMSE and LLSE predicted values are $E[Y]$.

Problem 8. Expected return of the uniform-sum program

Let $X_1, X_2, \dots$ be independent random variables, each uniformly distributed on $(0, 1)$. Define

$$S_n = \sum_{i=1}^n X_i, \quad N = \min\{n \geq 1 : S_n > 1\}.$$

Here, N is the number of loop iterations, and the program returns S_N .

For a positive integer-valued random variable N , the definition of expectation gives

$$\mathbb{E}[N] = \Pr(N = 1) + 2\Pr(N = 2) + 3\Pr(N = 3) + \dots$$

Splitting each coefficient into a sum of ones and regrouping the nonnegative terms, we obtain

$$\begin{aligned}
 \mathbb{E}[N] &= [\Pr(N = 1) + \Pr(N = 2) + \Pr(N = 3) + \cdots] \\
 &\quad + [\Pr(N = 2) + \Pr(N = 3) + \cdots] \\
 &\quad + [\Pr(N = 3) + \cdots] + \cdots \\
 &= \sum_{i=1}^{\infty} \Pr(N \geq i) \\
 &= \sum_{n=0}^{\infty} \Pr(N > n).
 \end{aligned}$$

Since the increments are nonnegative,

$$\Pr(N > n) = \Pr(S_n \leq 1) = \frac{1}{n!},$$

where, for $n \geq 1$, the last equality follows from the volume of the region

$$\left\{ (x_1, \dots, x_n) : x_i \geq 0, \sum_{i=1}^n x_i \leq 1 \right\}.$$

For $n = 0$, the probability is $1 = 1/0!$. Therefore,

$$\mathbb{E}[N] = \sum_{n=0}^{\infty} \frac{1}{n!} = e.$$

We can derive the expected return value directly, without invoking Wald's equation. Define the actual contribution of the i th iteration by

$$Y_i = \begin{cases} X_i, & \text{if the } i\text{th iteration occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently,

$$Y_i = X_i \mathbf{1}_{\{N \geq i\}}.$$

The key observation is that whether the i th iteration occurs depends only on $X_1, \dots, X_{i-1}$, not on X_i . Hence the event $\{N \geq i\}$ is independent of X_i .

Since

$$\mathbb{E}[X_i] = \frac{1}{2},$$

we have

$$\mathbb{E}[Y_i] = \mathbb{E}[X_i] \Pr(N \geq i) = \frac{1}{2} \Pr(N \geq i).$$

The returned sum is

$$S_N = \sum_{i=1}^{\infty} Y_i.$$

All the terms are nonnegative, so we may interchange expectation and summation:

$$\begin{aligned}
 \mathbb{E}[S_N] &= \sum_{i=1}^{\infty} \mathbb{E}[Y_i] \\
 &= \frac{1}{2} \sum_{i=1}^{\infty} \Pr(N \geq i) \\
 &= \frac{1}{2} \mathbb{E}[N] \\
 &= \frac{e}{2} \approx 1.35914.
 \end{aligned}$$

Problem 9. Hoeffding's lemma

Suppose that $\mathbb{E}[X] = 0$ and $a \leq X \leq b$ almost surely. Then, for every $\lambda > 0$,

$$\mathbb{E}[e^{\lambda X}] \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

If $a = b$, the zero-mean assumption implies that $X = 0$ almost surely, so the result is immediate. Henceforth, assume that $a < b$.

Since $\mathbb{E}[X] = 0$ and $a \leq X \leq b$, we have $a \leq 0 \leq b$. For every $x \in [a, b]$,

$$x = \frac{b-x}{b-a}a + \frac{x-a}{b-a}b.$$

The two coefficients are nonnegative and sum to one. Therefore, by the convexity of $x \mapsto e^{\lambda x}$,

$$e^{\lambda x} \leq \frac{b-x}{b-a}e^{\lambda a} + \frac{x-a}{b-a}e^{\lambda b}.$$

Substituting $x = X$, taking expectations, and using $\mathbb{E}[X] = 0$, we obtain

$$\mathbb{E}[e^{\lambda X}] \leq \frac{b}{b-a}e^{\lambda a} + \frac{-a}{b-a}e^{\lambda b}.$$

Let

$$p = \frac{-a}{b-a} \in [0, 1], \quad t = \lambda(b-a) > 0.$$

Then

$$1-p = \frac{b}{b-a}, \quad \lambda a = -pt, \quad \lambda b = (1-p)t.$$

Consequently,

$$\mathbb{E}[e^{\lambda X}] \leq (1-p)e^{-pt} + pe^{(1-p)t}.$$

Define

$$h(t) = \log\left((1-p)e^{-pt} + pe^{(1-p)t}\right) = -pt + \log(1-p + pe^t).$$

It suffices to prove that

$$h(t) \leq \frac{t^2}{8}.$$

First, observe that

$$h(0) = 0.$$

Differentiating once gives

$$h'(t) = -p + \frac{pe^t}{1 - p + pe^t},$$

and hence

$$h'(0) = 0.$$

Differentiating again gives

$$h''(t) = \frac{p(1-p)e^t}{(1-p+pe^t)^2}.$$

Let

$$q(t) = \frac{pe^t}{1-p+pe^t} \in [0, 1].$$

Then

$$h''(t) = q(t)(1-q(t)).$$

For every $q \in [0, 1]$,

$$q(1-q) = \frac{1}{4} - \left(q - \frac{1}{2}\right)^2 \leq \frac{1}{4}.$$

Therefore,

$$h''(t) \leq \frac{1}{4}.$$

Using $h'(0) = 0$, for every $s \geq 0$ we have

$$h'(s) = \int_0^s h''(u) du \leq \frac{s}{4}.$$

Using $h(0) = 0$ and integrating once more, we obtain

$$h(t) = \int_0^t h'(s) ds \leq \int_0^t \frac{s}{4} ds = \frac{t^2}{8}.$$

It follows that

$$\mathbb{E}[e^{\lambda X}] \leq e^{h(t)} \leq e^{t^2/8} = \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Problem 10. Utility, Fairness, and RL Rewards

We use the feasible set

$$F = \{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2 : x_1 + 2x_2 \leq 12\}.$$

For logarithmic utility, both rates must be positive.

(a) Utility Functions and Allocation

A utility function represents preferences by

$$a \succeq b \iff u(a) \geq u(b).$$

Both $u(x) = x$ and $v(x) = \log x$ are strictly increasing, so both express “more service is better.” However, maximizing total utility under equal weights can give different answers.

For

$$\max_F (x_1 + x_2),$$

the optimum is

$$(12, 0),$$

because

$$x_1 + x_2 \leq x_1 + 2x_2 \leq 12.$$

For logarithmic utility,

$$\max_F (\log x_1 + \log x_2) = \max_F \log(x_1 x_2),$$

the optimum is

$$(6, 3).$$

Indeed, by AM–GM,

$$x_1 x_2 = \frac{1}{2} x_1 (2x_2) \leq \frac{1}{2} \left(\frac{x_1 + 2x_2}{2} \right)^2 \leq \frac{1}{2} \left(\frac{12}{2} \right)^2 = 18.$$

Equality requires

$$x_1 = 2x_2, \quad x_1 + 2x_2 = 12,$$

so $x_1 = 6$, $x_2 = 3$.

The reason is that the two utility functions have different marginal utilities:

$$u'(x) = 1, \quad v'(x) = \frac{1}{x}.$$

Knowing only that each user wants more service does not tell us whose extra service should count more. The utility functions and weights make that choice.

(b) Concavity and Transfers

The problem assumes

$$u'(x) > 0, \quad u''(x) < 0.$$

The first inequality means utility increases. The second means its slope decreases, so the curve becomes flatter as service increases.

Let

$$0 < a < b, \quad 0 < \delta < \frac{b-a}{2}.$$

Define

$$h(t) = u(a+t) + u(b-t), \quad 0 \leq t \leq \delta.$$

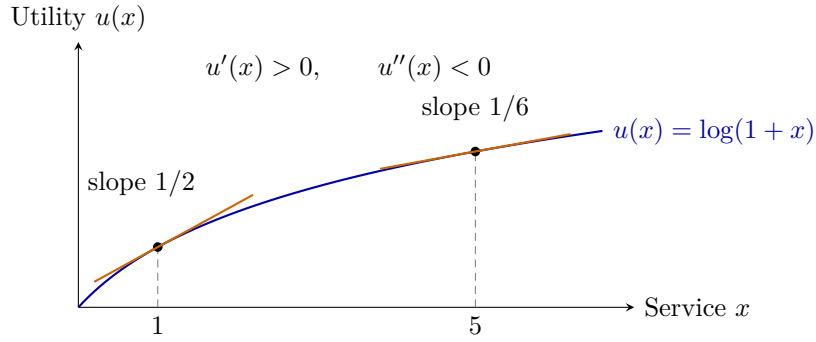


Figure 1: The curve rises but gets flatter: extra service adds less utility when a user already has more.

Then

$$h'(t) = u'(a+t) - u'(b-t).$$

Since

$$a+t < b-t$$

and u' is decreasing, we have

$$u'(a+t) > u'(b-t).$$

Therefore

$$h'(t) > 0.$$

Hence

$$h(\delta) > h(0),$$

which is exactly

$$u(a+\delta) + u(b-\delta) > u(a) + u(b).$$

In words, transferring a small amount of service from the user with more service to the user with less service increases total utility.

However, the best allocation need not be equal. The transfer above keeps total service $a+b$ fixed, but it may change resource consumption. For example, at $(6, 3)$, transferring service from user 1 to user 2 gives

$$(6-\delta) + 2(3+\delta) = 12 + \delta > 12,$$

which violates the constraint. Even with the strictly concave utility $\log x$, the best allocation is $(6, 3)$, not equal rates.

(c) Three Allocation Problems and Proportional Fairness

1. Maximize total service.

$$\max_F (x_1 + x_2).$$

Since

$$x_1 + x_2 \leq x_1 + 2x_2 \leq 12,$$

the maximum is 12, attained at

$$(12, 0).$$

2. Maximize logarithmic utility.

$$\max_F (\log x_1 + \log x_2).$$

This is equivalent to maximizing $x_1 x_2$. By AM–GM,

$$x_1 x_2 = \frac{1}{2} x_1 (2x_2) \leq \frac{1}{2} \left(\frac{x_1 + 2x_2}{2} \right)^2 \leq 18.$$

Equality requires

$$x_1 = 2x_2, \quad x_1 + 2x_2 = 12,$$

so

$$(x_1, x_2) = (6, 3).$$

3. Maximize minimum service. Let

$$m = \min\{x_1, x_2\}.$$

Then

$$x_1 \geq m, \quad x_2 \geq m,$$

so

$$x_1 + 2x_2 \geq 3m.$$

Since

$$x_1 + 2x_2 \leq 12,$$

we get

$$m \leq 4.$$

Equality occurs at

$$x_1 = x_2 = 4.$$

Thus the optimum is

$$(4, 4).$$

The three allocations are:

$$(12, 0), \quad (6, 3), \quad (4, 4).$$

They correspond respectively to maximum total service, proportional fairness, and maximum minimum service. The third objective gives equal rates in this particular feasible set.

Proof of proportional fairness. Let $C \subseteq \mathbb{R}_{>0}^n$ be convex, and let $\mathbf{x}^*$ maximize

$$W(\mathbf{x}) = \sum_{i=1}^n \log x_i.$$

For any $\mathbf{y} \in C$, convexity implies

$$\mathbf{x}(t) = \mathbf{x}^* + t(\mathbf{y} - \mathbf{x}^*) \in C, \quad 0 \leq t \leq 1.$$

Set

$$g(t) = W(\mathbf{x}(t)).$$

Since $\mathbf{x}^*$ is optimal,

$$g(t) \leq g(0).$$

Therefore its right derivative at zero cannot be positive:

$$g'(0^+) \leq 0.$$

A direct computation gives

$$g'(0^+) = \sum_{i=1}^n \frac{y_i - x_i^*}{x_i^*}.$$

Hence

$$\sum_{i=1}^n \frac{y_i - x_i^*}{x_i^*} \leq 0.$$

This says that when moving from the optimal allocation $\mathbf{x}^*$ to any feasible allocation $\mathbf{y}$, the sum of relative changes cannot be positive. In other words, the sum of percentage gains cannot exceed the sum of percentage losses. This is proportional fairness.

Weighting user 1 twice. Maximizing

$$2 \log x_1 + \log x_2$$

is equivalent to maximizing

$$x_1^2 x_2.$$

By AM–GM,

$$x_1^2 x_2 = 2 \left(\frac{x_1}{2} \right) \left(\frac{x_1}{2} \right) (2x_2) \leq 2 \left(\frac{x_1 + 2x_2}{3} \right)^3 \leq 2 \left(\frac{12}{3} \right)^3 = 128.$$

Equality requires

$$\frac{x_1}{2} = 2x_2, \quad x_1 + 2x_2 = 12.$$

Thus

$$x_1 = 8, \quad x_2 = 2.$$

The weight 2 means that user 1's utility counts twice in the objective, not that user 1 must receive twice as much service.

(d) Concavity under Uncertainty: Risk Aversion

Option A: receive 5 for sure.

Option B: receive 1 with probability 1/2, and 9 with probability 1/2.

Expected payoffs:

$$\mathbb{E}[X_A] = 5, \quad \mathbb{E}[X_B] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 9 = 5.$$

So a person maximizing expected payoff values A and B equally.

With logarithmic utility,

$$\mathbb{E}[\log X_A] = \log 5,$$

and

$$\mathbb{E}[\log X_B] = \frac{1}{2} \log 1 + \frac{1}{2} \log 9 = \log 3.$$

Since

$$\log 5 > \log 3,$$

A is better under logarithmic utility.

The certainty equivalent of B is the sure amount c satisfying

$$\log c = \mathbb{E}[\log X_B] = \log 3.$$

Thus

$$c = 3.$$

So B is equivalent to receiving 3 for sure.

Jensen's inequality: if u is concave, then

$$\mathbb{E}[u(X)] \leq u(\mathbb{E}[X]).$$

Thus receiving the mean payoff for sure is at least as good as taking the gamble. This is risk aversion.

The difference from part (b):

- In (d), one person compares uncertain outcomes. This is about risk.
- In (b), service is allocated across different people. This is about fairness.

(e) von Neumann–Morgenstern Expected Utility Theorem

Let the finite set of outcomes be

$$X = \{x_1, \dots, x_n\}.$$

A lottery is a probability vector

$$p = (p_1, \dots, p_n), \quad p_i \geq 0, \quad \sum_{i=1}^n p_i = 1.$$

Here $p \succeq q$ means p is at least as good as q , and $p \sim q$ means they are equally good.

The VNM theorem uses four axioms:

1. **Completeness:** For any two lotteries p, q ,

$$p \succeq q \quad \text{or} \quad q \succeq p$$

(or both).

2. **Transitivity:** If

$$p \succeq q, \quad q \succeq r,$$

then

$$p \succeq r.$$

3. **Continuity:** If

$$p \succ q \succ r,$$

then there exists $\alpha \in (0, 1)$ such that

$$q \sim \alpha p + (1 - \alpha)r.$$

4. **Independence:** For any lottery r and any $\alpha \in (0, 1)$,

$$p \succeq q \iff \alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r.$$

Mixing both choices with the same lottery in the same proportion keeps their preference order.

If these axioms hold, then there exists a utility function u such that lotteries are ranked by expected utility:

$$U(p) = \sum_{i=1}^n p_i u(x_i),$$

and

$$p \succeq q \iff U(p) \geq U(q).$$

Thus the decision maker chooses the lottery with the highest expected utility.

The utility representation is unique up to a positive affine transformation:

$$v(x) = au(x) + b, \quad a > 0.$$

Indeed,

$$\mathbb{E}[v] = a\mathbb{E}[u] + b,$$

and since $a > 0$, the ranking is unchanged.

The theorem does not require u to be concave or logarithmic, and it does not determine fairness across different users.

Using part (d): both $u(x) = x$ and $v(x) = \log x$ rank deterministic outcomes as

$$9 > 5 > 1.$$

But for the lotteries A and B,

$$\mathbb{E}[X_A] = 5 = \mathbb{E}[X_B],$$

so linear utility makes A and B equally good. Under logarithmic utility,

$$\mathbb{E}[\log X_A] = \log 5 > \log 3 = \mathbb{E}[\log X_B],$$

so A is better. Therefore a VNM utility cannot be subjected to an arbitrary strictly increasing transformation; only positive affine transformations preserve the ranking of lotteries.

(f) From Utility to Reinforcement Learning Rewards

Reward hypothesis. The reward hypothesis in reinforcement learning says that an agent's goal can be described by a scalar reward, and the agent maximizes expected total reward:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right].$$

This is similar to VNM expected utility: both express a goal numerically and then maximize an expectation. However, VNM alone does not automatically give a stationary Markov reward function

$$r(s, a, s').$$

If the objective depends on history, the state may need to be augmented before the reward can be written as a Markov reward.

Two fairness objectives. Objective 1: compute fairness at every time step:

$$J_{\text{instant}}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} (\log X_1(t) + \log X_2(t)) \right].$$

Objective 2: first average each user's service over the whole horizon, then compute fairness:

$$J_{\text{average}}(\pi) = \mathbb{E}_\pi \left[\log \bar{X}_1 + \log \bar{X}_2 \right],$$

where

$$\bar{X}_i = \frac{1}{T} \sum_{t=0}^{T-1} X_i(t).$$

Take $T = 2$.

Strategy π_A :

$$(6, 3), \quad (6, 3).$$

Strategy π_B :

$$(10, 1), \quad (2, 5).$$

Compute:

$$J_{\text{instant}}(\pi_A) = 2(\log 6 + \log 3) = \log 324.$$

$$J_{\text{instant}}(\pi_B) = \log 10 + \log 1 + \log 2 + \log 5 = \log 100.$$

So J_{instant} prefers π_A .

For average service:

$$\pi_A : \quad (6, 3),$$

and

$$\pi_B : \quad \left(\frac{10+2}{2}, \frac{1+5}{2} \right) = (6, 3).$$

Therefore

$$J_{\text{average}}(\pi_A) = J_{\text{average}}(\pi_B) = \log 6 + \log 3 = \log 18.$$

So J_{average} is indifferent between the two strategies.

Thus the two objectives are not equivalent. J_{instant} prefers π_A , while J_{average} gives a tie.

Constructing a reward for average service. Set

$$R_{t+1} = 0 \quad (0 \leq t < T - 1),$$

and at the final step set

$$R_T = \log \bar{X}_1 + \log \bar{X}_2.$$

Then

$$\mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right] = \mathbb{E}_\pi [\log \bar{X}_1 + \log \bar{X}_2] = J_{\text{average}}(\pi).$$

To make this reward Markov with respect to an augmented state, keep track of cumulative service:

$$C_i(0) = 0, \quad C_i(t+1) = C_i(t) + X_i(t).$$

Define the augmented state

$$\tilde{S}_t = (S_t, C_1(t), C_2(t), t).$$

Then

$$\bar{X}_i = \frac{C_i(T)}{T},$$

and the clock t tells us when to pay the terminal reward. If S_t already describes the environment's Markov dynamics, this gives a fixed Markov reward rule on the augmented state: pay zero except on a transition ending at time T , when the reward uses the new totals.

(g) Connection between VNM Expected Utility and RL Reward Hypothesis

VNM expected utility theory says that, under the four axioms, lottery preferences can be ranked by expected utility:

$$p \succeq q \iff \mathbb{E}_p[u(X)] \geq \mathbb{E}_q[u(X)].$$

The RL reward hypothesis applies a similar idea to sequential decisions: policies are ranked by expected total reward:

$$\pi \succeq \pi' \iff \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right] \geq \mathbb{E}_{\pi'} \left[\sum_{t=0}^{T-1} R_{t+1} \right].$$

For example, in part (f), to optimize average-service fairness, set all earlier rewards to zero and set the final reward to

$$R_T = \log \bar{X}_1 + \log \bar{X}_2.$$

Then maximizing expected total reward is exactly maximizing J_{average} .

However, VNM alone does not automatically tell us:

- what the state should contain;
- how each step's reward should be defined;
- whether the resulting reward is Markov.

VNM is a theorem based on preference axioms. The RL reward hypothesis is a related idea about representing goals in sequential decisions, not a consequence automatically guaranteed by VNM.